Chapter 32: Digestion and Nutrition

Feeding Mechanisms

- Most animals are **heterotrophic** organisms that depend on already synthesized organic compounds of plants and other animals.
- Animals are categorized by dietary habits:
  - **Herbivorous**: Feed mainly on plant life.
  - **Carnivorous**: Feed mainly on herbivores and other carnivores.
  - **Omnivorous**: Eat both plants and animals.
  - **Saprophagous**: Feed on decaying organic matter.

Feeding on Particulate Matter

- **Suspension-feeding**: Using ciliated surfaces to produce currents that draw drifting food particles into the mouth.
  - Examples: Tube-dwelling polychaete worms, bivalve molluscs, hemichordates.

- **Filter-feeding**: A form of suspension feeding where animals possess filtering devices that strain food from water.
  - Examples: Microcrustaceans, herring, baleen whales.
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Figure 1: Some suspension and filter feeders and their feeding mechanisms. A, Marine fan worms (class Polychaeta) use ciliated tentacles. B, Bivalve molluscs use gills. C, Barnacles use thoracic appendages (cirri). D, Herring use gill rakers. E, Whalebone whales use baleen plates.

  - Figure 32.1 illustrates various suspension and filter feeders:
    - (A) Marine fan worms use ciliated tentacles.
    - (B) Bivalve molluscs use gills to trap particles in mucus.
    - (C) Barnacles sweep thoracic appendages (cirri) to trap plankton.
    - (D) Herring use gill rakers to strain plankton.
    - (E) Whalebone whales use baleen plates to filter krill.

- **Deposit-feeding**: Exploiting deposits of disintegrated organic material (detritus) on and in the substratum.

- Examples: Annelids, scaphopod molluscs.
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  Figure 2: The annelid **Amphitrite** is a deposit feeder that lives in a mucus-lined burrow and extends long feeding tentacles in all directions across the surface. Food trapped on mucus is conveyed along the tentacles to the mouth.
- Figure 32.2 shows the annelid **Amphitrite**, a deposit feeder that extends long tentacles to gather food particles trapped in mucus.

Feeding on Food Masses

- Predators must locate, capture, hold, and swallow prey.
- Adaptations include teeth, beaks, and jaws.
  - **Nereis** (polychaete) has a muscular pharynx with chitinous jaws.
  - Snakes have recurved teeth and distensible jaws for large meals.
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    Figure 3: The African egg-eating snake, **Dasypeltis**, subsists entirely on hard-shelled birds' eggs. Adaptations include reduced dentition, expansible jaws, and vertebral spurs to puncture the shell.
  - Figure 32.3 shows the egg-eating snake **Dasypeltis**, which swallows eggs whole and punctures them with vertebral spurs.
- **Mastication**: Chewing or crushing of food, found mainly in mammals.
- Mammalian teeth types:
  - **Incisors**: Biting, cutting, stripping.
  - **Canines**: Seizing, piercing, tearing.
  - **Premolars** and **Molars**: Grinding and crushing.

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  Figure 4: Structure of human molar tooth. Enamel covers the crown; dentine composes the mass; cementum covers the root. The pulp cavity contains nerves and blood vessels.
- Figure 32.4 details the structure of a human molar, showing enamel, dentine, pulp cavity, and cementum.

- Dentition is modified based on diet:
  - Herbivores often lack canines but have well-developed molars.
  - Rodents have self-sharpening incisors that grow continuously.
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    Figure 5: Mammalian dentition. A, Gray fox (carnivore) with four tooth types. B, Woodchuck (rodent) with chisel-like incisors. C, White-tailed deer (herbivore) with grinding molars.
  - Figure 32.5 compares dentition: (A) Carnivore (fox) has all four types; (B) Rodent (woodchuck) has chisel-like incisors; (C) Herbivore (deer) has grinding molars.

- Specialized teeth:
  - Elephant tusks are modified upper incisors.

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Figure 6: An African elephant loosening soil from a salt lick with its tusk, a modified upper incisor used for defense, attack, and rooting.
- Figure 32.6 shows an elephant using its tusk for digging.

- Herbivores use **cellulase**-producing microorganisms (bacteria and protozoa) in their gut to digest cellulose, as no metazoan produces intestinal cellulase.

Feeding on Fluids

- **Fluid-feeding**: Characteristic of parasites (endoparasites and ectoparasites) but also some free-living forms.
- Examples: Leeches, lampreys, mosquitoes, ticks, mites.
- Mosquitoes have needlelike mouthparts to puncture skin and inject anticoagulant saliva.

Digestion

- **Digestion**: Mechanical and chemical breakdown of organic foods into small units for absorption.

- **Intracellular digestion**: Occurs within a cell.
  - Food particles are enclosed in a food vacuole by **phagocytosis**.
  - Digestive enzymes from lysosomes are added.
  - Limited to particles small enough to be phagocytized.
  - Found in protozoa and sponges.
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Figure 7: Intracellular digestion. Lysosomes containing digestive enzymes fuse with food vacuoles formed by phagocytosis. Usable products are absorbed, and wastes are expelled by exocytosis.

  - Figure 32.7 illustrates intracellular digestion: Lysosomes fuse with food vacuoles, enzymes digest food, and wastes are extruded by exocytosis.
- **Extracellular digestion**: Occurs in the lumen of an **alimentary canal**.
  - Cells lining the lumen secrete enzymes; others absorb nutrients.
  - Allows digestion of large food masses.
  - Characteristic of arthropods and vertebrates.

Action of Digestive Enzymes

- Digestive enzymes are **hydrolytic enzymes** (**hydrolases**).
- They break chemical bonds by **hydrolysis** (adding water): $\mathrm{R^{-}R} + \mathrm{H_2O} \longrightarrow \mathrm{R^{-}OH} + \mathrm{R^{-}H}$.
- Specific enzymes for each class of organic compounds:
  - Proteins -> Amino acids.
  - Carbohydrates -> Simple sugars.
  - Fats -> Glycerol, fatty acids, monoglycerides.

Motility in Alimentary Canal

- Food is moved by **cilia** (acoelomates, pseudocoelomates) or **musculature** (coelomates).
- Gut musculature typically has longitudinal and circular layers.
- **Segmentation**: Alternate constriction of rings of smooth muscle to mix food.
- **Peristalsis**: Waves of contraction behind the food mass (**bolus**) to propel it forward.
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Figure 8: Movement of intestinal contents. A, Segmentation mixes food with enzymes. B, Peristalsis propels food forward by waves of contraction.
- Figure 32.8 depicts (A) Segmentation mixing food and (B) Peristalsis moving food.

Organization and Regional Function of Alimentary Canals

- Five major regions:
  1. Reception
  2. Conduction and Storage
  3. Grinding and Early Digestion
  4. Terminal Digestion and Absorption
  5. Water Absorption and Concentration of Solids

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Figure 9: Generalized digestive tracts of a vertebrate and an insect, showing major functional regions: reception, conduction/storage, grinding/early digestion, terminal digestion/absorption, and water absorption.

- Figure 32.9 compares generalized vertebrate and insect digestive tracts, highlighting functional regions.

Receiving Region

- Includes mouthparts, buccal cavity, and pharynx.
- **Salivary glands** secrete mucus for lubrication and enzymes.
  - **Salivary amylase**: Begins hydrolysis of starch into **maltose**. Found in primates, some herbivores.
- **Tongue**: Vertebrate innovation for food manipulation, swallowing, and chemosensation (taste buds).
- Swallowing in humans:
  - Tongue pushes food to pharynx.
  - Soft palate closes nasal cavity.
  - **Epiglottis** closes trachea.
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Figure 10: Oral cavity and swallowing in humans. A, Sagittal section. B-D, Sequence of swallowing: soft palate closes nasal cavity, epiglottis closes trachea, and esophageal muscles propel food.

- Figure 32.10 shows the anatomy of the oral cavity and the sequence of swallowing events preventing food from entering the trachea.

Conduction and Storage Region

- **Esophagus**: Transfers food to digestive region.
- **Crop**: Expanded region of esophagus for storage (invertebrates, birds).
  - In birds, softens food or allows mild fermentation.

Region of Grinding and Early Digestion

- **Stomach**: Provides initial digestion, storage, and mixing.
- **Gizzard**: Muscular organ for grinding (birds, oligochaetes).
- **Proventriculus**: Grinding organ in insects.
- **Digestive diverticula**: Blind tubules for secretion/absorption (e.g., ceca in polychaetes, pyloric ceca in sea stars).
- Vertebrate Stomach:
  - **Cardiac sphincter**: Allows food entry, prevents regurgitation.
  - **Pyloric sphincter**: Regulates flow to intestine.
  - **Gastric juice** contains:
    - **Mucus**: Protects stomach lining (secreted by goblet cells).
    - **Pepsinogen**: Precursor to **pepsin** (secreted by chief cells).
    - **Hydrochloric acid (HCl)**: Activates pepsinogen to pepsin (secreted by parietal/oxyntic cells).
  - **Pepsin**: Protease that splits large proteins into polypeptides; active at pH 1.6-2.4.
  - **Rennin**: Milk-curdling enzyme in ruminants.
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Figure 11: Dr. William Beaumont collecting gastric juice from Alexis St. Martin, who had a permanent gastric fistula.

  - Figure 32.11 shows Dr. William Beaumont collecting gastric juice from Alexis St. Martin, a classic study in digestion.

Region of Terminal Digestion and Absorption: Intestine

- **Intestine**: Site of digestion and absorption.
- Strategies to increase surface area:
  - Coiling (mammals).
  - **Typhlosole** (oligochaetes).
  - Spiral valve (sharks).
  - **Villi**: Fingerlike projections (birds, mammals).
  - **Microvilli**: Processes on individual cells.

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Figure 12: Scanning electron micrograph of rat intestine showing villi that vastly increase absorptive surface area.

- Figure 32.12 is a scanning electron micrograph of rat intestine villi.
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Figure 13: Organization of vertebrate digestive tract. A, Layers: mucosa, submucosa, muscle, serosa. B, Mucosal lining with villi. C, Single mucosal cell. D, Microvilli on mucosal cell surface.

- Figure 32.13 details the layers of the vertebrate digestive tract (mucosa, submucosa, muscle, serosa) and the structure of villi and microvilli.

Digestion in the Vertebrate Small Intestine

- **Duodenum**: Initial segment where **pancreatic juice** and **bile** enter.
- pH is raised from 1.5 to 7 by bicarbonate.
- **Bile**:
  - Produced by **liver**, stored in **gallbladder**.
  - Contains **bile salts** (emulsify fats) and **bile pigments** (breakdown of hemoglobin).
  - No enzymes.
- **Pancreatic Juice**:
  - **Trypsin** and **Chymotrypsin**: Proteases.
    - Hydrolysis of a peptide linkage: $\mathrm{R{-}CO{-}NH{-}R'} + \mathrm{H_2O} \longrightarrow \mathrm{R{-}COOH} + \mathrm{H_2N{-}R'}$.
  - **Carboxypeptidase**: Removes amino acids from carboxyl ends.

- **Pancreatic lipase**: Hydrolyzes fats.
- **Pancreatic amylase**: Hydrolyzes starch.
- **Nucleases**: Degrade RNA and DNA.
- **Membrane Enzymes** (on microvilli):
  - **Aminopeptidase**: Splits terminal amino acids.
  - **Disaccharidases**: **Maltase** (maltose -> glucose), **Sucrase** (sucrose -> fructose + glucose), **Lactase** (lactose -> glucose + galactose).
  - **Alkaline phosphatase**, **Nucleotidases**, **Nucleosidases**.
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Figure 14: Secretions of a mammalian alimentary canal (saliva, gastric juice, bile, pancreatic juice, membrane enzymes) with principal components and pH.
- Figure 32.14 summarizes secretions, their composition, and pH in the mammalian alimentary canal.

Absorption
- Carbohydrates absorbed as monosaccharides (active transport).
- Proteins absorbed as amino acids (active transport).
- Fats:
  - Emulsified by bile salts.
  - Digested to fatty acids and monoglycerides.
  - Form **micelles**.
  - Absorbed by diffusion, resynthesized to triglycerides, enter **lacteals** (lymphatic system).

Region of Water Absorption and Concentration of Solids
- **Large Intestine (Colon)**: Reabsorbs water and ions.
- Consolidates indigestible remnants into feces.
- Contains bacteria that synthesize vitamins (K and B complex).
- Insects have rectal glands for water absorption.
- Reptiles and birds absorb water in the cloaca (uric acid excretion).

Regulation of Food Intake
- Regulated by **hunger centers** in the hypothalamus and brain stem.
- **Obesity**: BMI >= 30.

- **Brown fat**: Specialized adipose tissue for heat generation (**diet-induced thermogenesis**).
  - Contains **uncoupling protein** in mitochondria (generates heat instead of ATP).
- **Leptin**: Hormone produced by fat cells; acts on hypothalamus to diminish appetite and increase thermogenesis.

Regulation of Digestion

- Coordinated by gastrointestinal (GI) hormones.
- **Gastrin**:
  - Secreted by stomach (pyloric region).
  - Stimulus: Protein food, parasympathetic nerves.
  - Action: Stimulates HCl secretion and gastric motility.
- **Cholecystokinin (CCK)**:
  - Secreted by upper small intestine.
  - Stimulus: Fatty acids and amino acids.
  - Action: Stimulates gallbladder contraction (bile release), pancreatic enzyme secretion, and satiety.
- **Secretin**:
  - Secreted by duodenal wall.
  - Stimulus: Acid in stomach/intestine.
  - Action: Stimulates alkaline pancreatic fluid (bicarbonate), inhibits gastric motility.
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Figure 15: Three hormones of digestion: Gastrin (stimulates acid), CCK (stimulates bile/pancreatic secretion, satiety), and Secretin (stimulates bicarbonate, inhibits motility).

- Figure 32.15 illustrates the actions and feedback loops of Gastrin, CCK, and Secretin.

Nutritional Requirements

- Requirements: Carbohydrates, proteins, fats, water, mineral salts, vitamins.
- **Vitamins**: Simple organic compounds required in small amounts; function as coenzymes.
  - **Water-soluble**: B complex, C.
  - **Fat-soluble**: A, D, E, K.
- **Essential nutrients**: Must be supplied in diet (cannot be synthesized).

- Humans require 30 organic compounds and 21 elements.

- **Lipids**: Energy source. Essential fatty acids (e.g., linoleic) required.
- **Proteins**: Source of amino acids.
  - 8-10 **essential amino acids** for humans.
  - **Protein complementarity**: Mixing plant sources to get all essential amino acids (e.g., grains + legumes).
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Figure 16: Human Nutrient Requirements, including water-soluble vitamins, fat-soluble vitamins, minerals (major and trace), amino acids, and polyunsaturated fatty acids.

- Table 32.1 lists human nutrient requirements.
- Malnutrition:
  - **Marasmus**: General undernourishment (low calories and protein).
  - **Kwashiorkor**: Protein malnourishment (adequate calories, low protein).
  - Severe malnutrition in infants leads to reduced brain cell number and permanent damage.
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Figure 17: Effect of early malnutrition on cell number in human brain. Malnourished infants have fewer brain cells than normal infants.

  - Figure 32.16 shows the effect of malnutrition on brain cell DNA content.

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Figure 18: Refugee child suffering severe malnutrition.

- Figure 32.17 depicts a child suffering from severe malnutrition.